

Accelerating Convergence to Equilibrium in Gaussian Diffusions

Optimal Rates and Applications to Langevin Dynamics

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A Warm-Up Example

$$\underbrace{\mu = \mathcal{N}(0, \text{diag}(9, 1))}_{\text{same target}}$$

$$\underbrace{\text{tr}(\mathbf{D}) = 1}_{\text{same noise budget}}$$

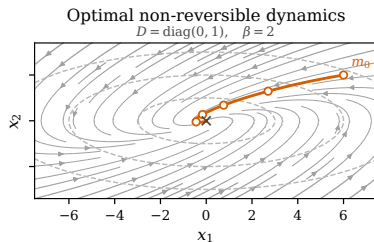
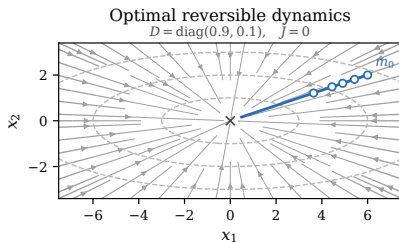


Figure: Mean drift trajectories with asymptotic rates 0.1 and 0.5.

- Where does the factor of five come from?

The Design Problem

- Does the process preserve the target distribution?
- How quickly does it forget its initial state?

Research Question

Given a centred Gaussian target and a fixed noise budget, how should the **noise** and the non-reversible **drift** be chosen so that the process forgets its initial state as rapidly as possible?

coupling $\longrightarrow$ spectrum $\longrightarrow$ optimisation

Settings

- Target distribution:

$$\mu = \mathcal{N}(0, \mathbf{C}), \quad \mathbf{C} = \mathbf{C}^\top \succ 0.$$

Ornstein–Uhlenbeck diffusion:

$$dX_t = \sqrt{2\sigma} dB_t - \mathbf{A}X_t dt, \quad \mathbf{D} := \sigma\sigma^\top.$$

- The target is **invariant** if and only if

$$\mathbf{A}\mathbf{C} + \mathbf{C}\mathbf{A}^\top = 2\mathbf{D}.$$

$$\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{C}^{-1}, \quad \mathbf{D} \succeq 0, \quad \mathbf{J}^\top = -\mathbf{J}.$$

Definitions

- A **coupling** of probability measures μ and ν is a pair (X, Y) on a common probability space such that

$$X \sim \mu, \quad Y \sim \nu.$$

- The **L^2 -Wasserstein distance**:

$$\mathcal{W}_2(\mu, \nu) := \inf_{\substack{X \sim \mu \\ Y \sim \nu}} \left(\mathbb{E} \left[\|X - Y\|^2 \right] \right)^{\frac{1}{2}}.$$

- It measures how closely two laws can be placed on the same probability space.

Synchronous Coupling

- Start from x and y , but use the same Brownian motion:

$$dX_t^x = \sqrt{2}\sigma dB_t - \mathbf{A}X_t^x dt,$$

$$dX_t^y = \sqrt{2}\sigma dB_t - \mathbf{A}X_t^y dt.$$

- Their difference $\Delta_t := X_t^x - X_t^y$ satisfies

$$\frac{d}{dt}\Delta_t = -\mathbf{A}\Delta_t, \quad \Delta_t = e^{-t\mathbf{A}}(x - y).$$

- The trajectories are random; their separation is not.

Wasserstein Contraction Coefficients

- The transition distributions have the same covariance:

$$\begin{aligned}p_t(x, \cdot) &= \mathcal{N}(e^{-t\mathbf{A}}x, \Gamma_t), \\p_t(y, \cdot) &= \mathcal{N}(e^{-t\mathbf{A}}y, \Gamma_t).\end{aligned}$$

- The synchronous coupling is optimal, and

$$\mathcal{W}_2(p_t(x, \cdot), p_t(y, \cdot)) = \|e^{-t\mathbf{A}}(x - y)\|.$$

Hence the **contraction coefficient** is

$$\alpha_2(p_t) := \sup_{x \neq y} \frac{\mathcal{W}_2(p_t(x, \cdot), p_t(y, \cdot))}{\|x - y\|} = \|e^{-t\mathbf{A}}\|.$$

Wasserstein Contraction Rate

■ Define

$$\lambda^*(\mathbf{A}) := \min_{\lambda \in \text{spec}(\mathbf{A})} \text{Re } \lambda.$$

■ Then

$$\lim_{t \rightarrow \infty} \frac{1}{t} \log \alpha_2(p_t) = -\lambda^*(\mathbf{A}).$$

stochastic contraction problem $\rightarrow$ deterministic matrix problem

2-D Case

■ Write

$$\mathbf{C} = \begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} \varepsilon & 0 \\ 0 & \gamma \end{pmatrix}, \quad \mathbf{J} = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}.$$

Then

$$\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{C}^{-1} = \begin{pmatrix} \frac{\varepsilon}{u} & \frac{\beta}{v} \\ -\frac{\beta}{u} & \frac{\gamma}{v} \end{pmatrix}.$$

■ The eigenvalues of $\mathbf{A}$:

$$\lambda_{\pm} = \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right) \pm \sqrt{\frac{1}{4} \left(\frac{\varepsilon}{u} - \frac{\gamma}{v} \right)^2 - \frac{\beta^2}{uv}}.$$

Therefore,

$$\lambda^*(\varepsilon, \gamma, \beta) = \underbrace{\frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right)}_{\text{average spectral rate}} - \underbrace{\sqrt{\left[\frac{1}{4} \left(\frac{\varepsilon}{u} - \frac{\gamma}{v} \right)^2 - \frac{\beta^2}{uv} \right]_+}}_{\text{imbalance penalty}}.$$

2-D Case

- Since

$$\lambda_+ + \lambda_- = \text{tr}(\mathbf{A}) = \frac{\varepsilon}{u} + \frac{\gamma}{v},$$

the smaller real part cannot exceed their average:

$$\lambda^* \leq \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right).$$

- Equality holds whenever

$$|\beta| \geq \frac{1}{2} \left| \sqrt{\frac{v}{u}} \varepsilon - \sqrt{\frac{u}{v}} \gamma \right|.$$

$$\max_{\beta \in \mathbb{R}} \lambda^*(\varepsilon, \gamma, \beta) = \frac{1}{2} \left(\frac{\varepsilon}{u} + \frac{\gamma}{v} \right).$$

Total Noise Constraint

- To prevent a trivial rescaling of time, impose

$$\text{tr}(\mathbf{D}) = \varepsilon + \gamma \leq 1.$$

- The optimal non-reversible rate is $\frac{1}{2} \max \left\{ \frac{1}{u}, \frac{1}{v} \right\}$. Assume $u \geq v$. One optimal choice is

$$\varepsilon = 0, \quad \gamma = 1, \quad |\beta| \geq \frac{1}{2} \sqrt{\frac{u}{v}}.$$

$$\mathbf{D}^* = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \lambda_{\text{nr}}^* = \frac{1}{2v}.$$

- Noise enters only the smallest-variance direction; the drift transports its effect.

Reversible vs Non-Reversible

- In the reversible case, $\beta = 0$. Then

$$\lambda^*(\varepsilon, \gamma, 0) = \min \left\{ \frac{\varepsilon}{u}, \frac{\gamma}{v} \right\}.$$

The optimum balances the two coordinate rates:

$$\frac{\varepsilon}{u} = \frac{\gamma}{v}, \quad \varepsilon + \gamma = 1.$$

$$\lambda_{\text{rev}}^* = \frac{1}{u + v}.$$

- With $\kappa(\mathbf{C}) = \frac{\max\{u, v\}}{\min\{u, v\}}$,

$$\frac{\lambda_{\text{nr}}^*}{\lambda_{\text{rev}}^*} = \frac{1 + \kappa(\mathbf{C})}{2}.$$

- Isotropic targets give no gain; anisotropy creates the opportunity for acceleration.

Example

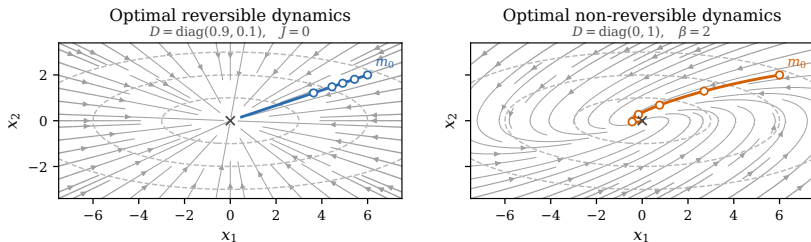


Figure: Reversible and non-reversible motions with $\lambda_{\text{rev}}^* = 0.1$ and $\lambda_{\text{nr}}^* = 0.5$.

- Reversible: balance two uncoupled rates.
- Non-reversible: transport the effect between coordinates.

d-D Case

- Let $\mathbf{H} := \mathbf{C}^{-1}$. For every admissible $\mathbf{J}$,

$$\sum_{k=1}^d \operatorname{Re} \lambda_k = \operatorname{tr}((\mathbf{D} + \mathbf{J})\mathbf{H}) = \operatorname{tr}(\mathbf{D}\mathbf{H}).$$

Therefore,

$$\lambda^* \leq \frac{1}{d} \operatorname{tr}(\mathbf{D}\mathbf{C}^{-1}).$$

- A separate linear-algebra construction chooses $\mathbf{J}$ so that all real parts equal this average:

$$\max_{\mathbf{J}^T = -\mathbf{J}} \lambda^*((\mathbf{D} + \mathbf{J})\mathbf{C}^{-1}) = \frac{1}{d} \operatorname{tr}(\mathbf{D}\mathbf{C}^{-1}).$$

d -D Case

- Under $\text{tr}(\mathbf{D}) \leq \eta$,

$$\text{tr}(\mathbf{D}\mathbf{C}^{-1}) \leq \eta \lambda_{\max}(\mathbf{C}^{-1}).$$

$$\max_{\mathbf{D}, \mathbf{J}} \lambda^* = \frac{\eta}{d} \lambda_{\max}(\mathbf{C}^{-1}).$$

- If $\mathbf{v}_{\max}$ is a unit eigenvector of $\mathbf{C}^{-1}$ for $\lambda_{\max}(\mathbf{C}^{-1})$, an optimiser is

$$\mathbf{D}^* = \eta \mathbf{v}_{\max} \mathbf{v}_{\max}^{\top}$$

together with a suitable antisymmetric matrix $\mathbf{J}^*$.

Langevin Dynamics

- Kinetic Langevin dynamics is

$$dX_t = V_t dt,$$

$$dV_t = -\nabla U(X_t) dt - \gamma V_t dt + \sqrt{2\gamma} dB_t.$$

- Its invariant distribution is

$$\mu(dx dv) \propto \exp\left(-U(x) - \frac{1}{2}\|v\|^2\right) dx dv.$$

- The noise reaches position through a deterministic relay:

$$B_t \longrightarrow V_t \longrightarrow X_t.$$

Langevin Dynamics

■ Let

$$U(x) = \frac{1}{2} x^\top \mathbf{C}^{-1} x,$$

and let $\sigma_1^2, \dots, \sigma_d^2$ be the eigenvalues of $\mathbf{C}$. Each position–velocity mode has eigenvalues

$$\lambda_{i,\pm} = \frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \frac{1}{\sigma_i^2}}.$$

■ The slowest mode is **optimally critically damped**:

$$\gamma^* = \frac{2}{\max_i \sigma_i} = \frac{2}{\sqrt{\varrho(\mathbf{C})}}, \quad \lambda^* = \frac{1}{\sqrt{\varrho(\mathbf{C})}}.$$

Beyond Gaussianity

- Suppose

$$K\mathbf{I}_d \preceq \nabla^2 U(x) \preceq L\mathbf{I}_d.$$

Under synchronous coupling, let

$$Z_t = X_t - \tilde{X}_t, \quad W_t = V_t - \tilde{V}_t.$$

- Their difference satisfies

$$\frac{d}{dt} \begin{pmatrix} Z_t \\ W_t \end{pmatrix} = - \underbrace{\begin{pmatrix} 0 & -\mathbf{I}_d \\ \mathbf{H}_t & \gamma \mathbf{I}_d \end{pmatrix}}_{\mathbf{A}_t} \begin{pmatrix} Z_t \\ W_t \end{pmatrix}, \quad K\mathbf{I}_d \preceq \mathbf{H}_t \preceq L\mathbf{I}_d.$$

- Unlike the Gaussian case, $\mathbf{A}_t$ changes along the coupled paths.

Beyond Gaussianity

- For $\gamma > \sqrt{L}$, define

$$\mathbf{G}_\gamma = \begin{pmatrix} \gamma^2 \mathbf{I}_d & \gamma \mathbf{I}_d \\ \gamma \mathbf{I}_d & 2\mathbf{I}_d \end{pmatrix}.$$

Then, for every $q \in [1, \infty)$,

$$\alpha_q(p_t) \leq \sqrt{\bar{\kappa}(\mathbf{G}_\gamma)} e^{-c(\gamma)t}, \quad c(\gamma) = \frac{\min\{K, \gamma^2 - L\}}{\gamma}.$$

- The rate bound is maximised at

$$\gamma^* = \sqrt{K + L}, \quad c(\gamma^*) = \frac{K}{\sqrt{K + L}}.$$

Limitations

Spectral acceleration is not yet algorithmic acceleration:

- λ^* is an asymptotic continuous-time slope, not a complete finite-time guarantee.
- A defective boundary eigenvalue may prevent a uniform estimate with exponent exactly λ^* .
- A large antisymmetric drift may create stiffness and require smaller discretisation steps.
- Continuous-time rates do not include discretisation error or computational cost.

Conclusion

- Invariance:

$$\mathbf{A} = (\mathbf{D} + \mathbf{J})\mathbf{C}^{-1}.$$

- Contraction:

$$\alpha_2(\mathbf{p}_t) = \|\mathbf{e}^{-t\mathbf{A}}\|.$$

- Optimisation under $\text{tr}(\mathbf{D}) \leq \eta$:

$$\lambda_{\text{opt}}^* = \frac{\eta}{d} \lambda_{\max}(\mathbf{C}^{-1}).$$

Take Home Message

Rank-one noise and antisymmetric transport remove the slow direction as a bottleneck. Faster convergence can come from arranging the motion, not from adding more noise.

Questions?

Langevin Dynamics

- Fix the one-dimensional quadratic potential

$$U(x) = \frac{k}{2}x^2, \quad k = \frac{1}{\sigma^2}, \quad X \sim \mathcal{N}(0, \sigma^2).$$

- Compare how noise reaches the target coordinate:

overdamped: $B_t \longrightarrow X_t$,

kinetic: $B_t \longrightarrow V_t \longrightarrow X_t$,

three-level: $B_t \longrightarrow Y_t \longrightarrow V_t \longrightarrow X_t$.

- The models have different dimensions and different noise budgets.

Overdamped Langevin

- The dynamics is

$$dX_t = -kX_t dt + \sqrt{2}dB_t.$$

- It preserves $\mathcal{N}(0, k^{-1})$ and has

$$\lambda_{\text{od}} = k = \frac{1}{\sigma^2}.$$

- Here the noise acts directly on the target coordinate.

Kinetic Dynamics

- The characteristic polynomial of the drift is

$$\lambda^2 - \gamma\lambda + k.$$

Hence

$$r_{kl}(\gamma) = \begin{cases} \frac{\gamma}{2}, & 0 < \gamma \leq 2\sqrt{k} \\ \frac{\gamma}{2} - \sqrt{\frac{\gamma^2}{4} - k}, & \gamma > 2\sqrt{k} \end{cases}.$$

- The two eigenvalues coalesce at critical damping:

$$\gamma_{kl}^* = 2\sqrt{k}, \quad \lambda_{kl}^* = \sqrt{k} = \frac{1}{\sigma}.$$

A Three-Level Langevin Lift

- Introduce an auxiliary variable Y_t :

$$dX_t = V_t dt,$$

$$dV_t = -kX_t dt + \alpha Y_t dt,$$

$$dY_t = -\alpha V_t dt - \gamma Y_t dt + \sqrt{2\gamma} dB_t.$$

- The characteristic polynomial is

$$p_{\alpha,\gamma}(\lambda) = \lambda^3 - \gamma\lambda^2 + (k + \alpha^2)\lambda - k\gamma.$$

- At the optimum,

$$p_{\alpha^*,\gamma^*}(\lambda) = \left(\lambda - \sqrt{3k}\right)^3.$$

Comparing coefficients gives

$$\alpha^* = \sqrt{8k}, \quad \gamma^* = 3\sqrt{3k}, \quad \lambda_{3L}^* = \sqrt{3k}.$$

Comparisons

Dynamics	Spectral rate	Noise budget $\eta = \text{tr}(\mathbf{D})$
Overdamped	k	1
Kinetic	$\sqrt{k}$	$2\sqrt{k}$
Three-level	$\sqrt{3k}$	$3\sqrt{3k}$

- Kinetic is faster than overdamped exactly when $\sigma > 1$.
- Three-level is faster than overdamped exactly when $\sigma > \frac{1}{\sqrt{3}}$.
- These continuous-time spectral rates do not constitute a fair algorithmic comparison.

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